

Friction and Circular Motion, **solved**

Step-by-step solutions to every practice problem in Chapter 4. Try each problem on your own before you read its solution.

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P4.1

Friction comes from temporary bonds at the microscopic contact points (asperities) where two rough surfaces touch.

When the box has been at rest for a while, the asperities have had time to settle into each other. Under the load, the contact spots also creep and flatten, so they grow larger and form more, stronger bonds. Breaking all of these at once to start the motion takes a relatively large force, the maximum static friction $\mu_s N$.

Once the box is sliding, each contact lasts only a very short time before it is torn apart, so fewer and weaker bonds form at any moment. The kinetic friction $\mu_k N$ is therefore smaller. That is why $\mu_k < \mu_s$, and why a box is easier to keep moving than to start.

P4.2

Examples where friction points along the motion:

1. **A crate on the bed of an accelerating truck.** The crate speeds up with the truck. The only horizontal force on it is static friction from the bed, so that friction must point **forward**, in the direction of motion.
2. **Walking or driving.** Your shoe pushes backward on the ground. Static friction from the ground pushes your foot, and therefore you, **forward**. The same is true for the driven wheels of a car.
3. A box on a conveyor belt that is speeding up.

What friction really opposes is *relative sliding*, or the tendency to slide, between the two surfaces in contact. It does not oppose the motion of a body relative to the ground. In example 1, without friction the crate would slide *backward* relative to the bed, so friction acts forward to prevent that.

P4.3

Yes, it is accelerating. Its velocity changes direction continuously, so it has a centripetal acceleration v^2/r towards the centre of the curve.

The centripetal force is provided by **static friction** from the road on the tyres, acting sideways towards the centre. On a banked road, part of it comes from the horizontal component of the normal force.

No centrifugal force acts in the ground frame. In the ground (inertial) frame, the net force on the car points **inward**. A “centrifugal force” appears only if you choose to analyse the problem in the car’s rotating frame, where it is a pseudo-force with no agent.

P4.4

At the top, both the track’s normal force N and gravity mg point down, towards the centre:

$$N + mg = \frac{mv^2}{R}.$$

The rider feels weightless when $N = 0$, that is, when

$$v = \sqrt{gR}.$$

Faster than this: $N = m(v^2/R - g) > 0$. The seat (the track) pushes **down** on the rider, towards the centre. The rider feels pressed *into* the seat, even while upside down. At high enough speed, the apparent weight can exceed their normal weight.

P4.5

Static coefficient. The block is on the point of slipping at 35° , so this is the angle of repose:

$$\mu_s = \tan 35^\circ = 0.700.$$

Kinetic coefficient. While sliding down at 35° :

$$ma = mg \sin \theta - \mu_k mg \cos \theta \Rightarrow \mu_k = \frac{g \sin \theta - a}{g \cos \theta}.$$

$$\mu_k = \frac{9.8(0.574) - 2.0}{9.8(0.819)} = \frac{5.62 - 2.0}{8.03} = 0.451.$$

Check. $\mu_k < \mu_s$, as expected. ✓ The mass of the block does not matter.

P4.6

(a) The only horizontal force on the crate is static friction, which can be at most $\mu_s mg$. So

$$a_{\max} = \frac{\mu_s mg}{m} = \mu_s g = 0.40(9.8) = 3.92 \text{ m/s}^2.$$

This is independent of the crate's mass.

(b) Braking at no more than 3.92 m/s^2 , from $v^2 = u^2 + 2as$:

$$s = \frac{u^2}{2a} = \frac{20^2}{2(3.92)} = 51.0 \text{ m}.$$

Evaluate. Here friction points **backward** on the crate during braking. Again it opposes the crate's tendency to slide relative to the bed, which would be forward.

P4.7

Forces. For the table block: tension T forward and kinetic friction $\mu_k m_1 g = 0.25(3.0)(9.8) = 7.35$ N backward. For the hanging block: weight $m_2 g = 19.6$ N down and T up.

Equations:

$$m_1 : T - 7.35 = 3.0 a, \quad m_2 : 19.6 - T = 2.0 a.$$

Add:

$$12.25 = 5.0 a \Rightarrow a = 2.45 \text{ m/s}^2.$$

$$T = 19.6 - 2.0(2.45) = 14.7 \text{ N}.$$

Check m_1 : $14.7 - 7.35 = 7.35 = 3.0(2.45)$. ✓

P4.8

With no friction, $\tan \theta = v^2/(gr)$, so

$$v = \sqrt{gr \tan \theta} = \sqrt{9.8(200) \tan 15^\circ} = \sqrt{1960(0.268)} = \sqrt{525} = 22.9 \text{ m/s.}$$

In km/h: $22.9 \times 3.6 = 82.5 \text{ km/h.}$

P4.9

Speed and centripetal acceleration:

$$v = \frac{2\pi R}{T} = \frac{2\pi(15)}{30} = 3.14 \text{ m/s,} \quad a_c = \frac{v^2}{R} = \frac{9.87}{15} = 0.658 \text{ m/s}^2.$$

Top. The centre is below the rider, so take down as positive: $mg - N = ma_c$.

$$N_{\text{top}} = m(g - a_c) = 60(9.8 - 0.658) = 549 \text{ N.}$$

Bottom. The centre is above the rider, so take up as positive: $N - mg = ma_c$.

$$N_{\text{bottom}} = m(g + a_c) = 60(10.46) = 628 \text{ N.}$$

The true weight is 588 N, so the rider feels about 7% lighter at the top and 7% heavier at the bottom.

P4.10

The string tension is the only horizontal force, so it provides the centripetal force: $T = mv^2/R$. At the breaking tension:

$$v_{\text{max}} = \sqrt{\frac{T_{\text{max}} R}{m}} = \sqrt{\frac{50(1.2)}{0.50}} = \sqrt{120} = 11.0 \text{ m/s.}$$

After the string breaks, no horizontal force acts. By the first law, the puck moves in a **straight line along the tangent** to the circle at the point where it broke, at constant speed. It does *not* fly radially outward.

P4.11

Set the centripetal acceleration equal to 5000 g :

$$\omega^2 r = 5000\text{ g} \Rightarrow \omega = \sqrt{\frac{5000(9.8)}{0.080}} = \sqrt{6.125 \times 10^5} = 783\text{ rad/s}.$$

Convert to revolutions per minute:

$$\text{rpm} = \omega \times \frac{60}{2\pi} = 783 \times 9.55 = 7.47 \times 10^3\text{ rpm}.$$

The centrifuge must spin at about 7500 rpm .

P4.12

(a) At terminal speed, drag balances weight: $kv_t^2 = mg$, so

$$k = \frac{mg}{v_t^2} = \frac{75(9.8)}{60^2} = \frac{735}{3600} = 0.204\text{ kg/m}.$$

(b) Since $v_t = \sqrt{mg/k}$, increasing k by a factor of 25 divides v_t by 5:

$$v'_t = \frac{60}{5} = 12\text{ m/s}.$$

(c) Just after an instant opening, the skydiver is still moving at 60 m/s :

$$D = 25k(60)^2 = 25 \times 735\text{ N} = 18\,400\text{ N}.$$

Net upward force: $18\,400 - 735 = 17\,600\text{ N}$, so the deceleration is

$$a = \frac{17\,600}{75} = 235\text{ m/s}^2 \approx 24\text{ g}.$$

Why open slowly. A 24 g jolt could injure the skydiver and overload the harness and canopy. Real parachutes use a *slider* that slows the canopy's inflation over 1–3 s, keeping the peak deceleration to a few g. This is the same idea as a crumple zone: spread the change in velocity over more time and distance.

P4.13

(a) **Moving together.** The upper block is accelerated only by static friction from the lower block, so its acceleration is at most

$$a_{\max} = \frac{\mu_s mg}{m} = \mu_s g = 0.30(9.8) = 2.94 \text{ m/s}^2.$$

The whole system then needs

$$F_{\max} = (M + m) a_{\max} = 5.0(2.94) = 14.7 \text{ N}.$$

(b) $F = 25 \text{ N} > 14.7 \text{ N}$, so the blocks **slip**, and kinetic friction $\mu_k mg = 0.25(1.0)(9.8) = 2.45 \text{ N}$ acts between them.

- **Upper block:** friction forward, $a_1 = 2.45/1.0 = 2.45 \text{ m/s}^2$.
- **Lower block:** the applied force forward, and the reaction to that friction backward:

$$a_2 = \frac{25 - 2.45}{4.0} = 5.64 \text{ m/s}^2.$$

Check. $a_2 > a_1$, so the lower block slides forward relative to the upper one. Friction on the upper block is therefore forward, as assumed. ✓

P4.14

Derivation. At the maximum speed, the car tends to slide *up* and *out*, so friction acts **down** the slope with $f = \mu_s N$. Resolve vertically and horizontally (towards the centre):

$$\text{Vertical: } N \cos \theta - \mu_s N \sin \theta = mg.$$

$$\text{Radial: } N \sin \theta + \mu_s N \cos \theta = \frac{mv^2}{r}.$$

Divide the second equation by the first:

$$\frac{v^2}{gr} = \frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta} = \frac{\tan \theta + \mu_s}{1 - \mu_s \tan \theta}. \checkmark$$

For $v_{\min}$, friction acts up the slope, which simply reverses the sign of μ_s .

Numbers. $gr = 588 \text{ m}^2/\text{s}^2$ and $\tan 20^\circ = 0.364$.

$$v_{\max}^2 = 588 \cdot \frac{0.364 + 0.25}{1 - 0.091} = 588(0.675) = 397 \Rightarrow v_{\max} = 19.9 \text{ m/s (72 km/h)}.$$

$$v_{\min}^2 = 588 \cdot \frac{0.364 - 0.25}{1 + 0.091} = 588(0.104) = 61.4 \Rightarrow v_{\min} = 7.8 \text{ m/s (28 km/h)}.$$

The design speed, with no friction needed, lies between them: $\sqrt{588(0.364)} = 14.6 \text{ m/s}$.

P4.15

FBD of a rider. Three forces act: the weight mg (down), the normal force N from the wall (horizontal, towards the axis), and static friction f_s from the wall (up).

- **Radial:** $N = m\omega^2 R$. The wall provides the centripetal force.
- **Vertical:** $f_s = mg$, with $f_s \leq \mu_s N$.

So the requirement is $mg \leq \mu_s m\omega^2 R$, giving

$$\omega_{\min} = \sqrt{\frac{g}{\mu_s R}} = \sqrt{\frac{9.8}{0.40(2.5)}} = \sqrt{9.8} = 3.13 \text{ rad/s.}$$

In rpm: $3.13 \times 60/2\pi = 29.9$ rpm, about 30 rpm.

Mass dependence: none. The mass cancels, so a child and an adult stay pinned at the same speed. (The friction coefficient of their clothing still matters.)

P4.16

(a) Newton's second law, with down positive:

$$m \frac{dv}{dt} = mg - bv.$$

Separate the variables:

$$\int_0^v \frac{dv}{g - (b/m)v} = \int_0^t dt \Rightarrow -\frac{m}{b} \ln \left(1 - \frac{bv}{mg} \right) = t.$$

So

$$v(t) = \frac{mg}{b} \left(1 - e^{-bt/m} \right) = v_t \left(1 - e^{-t/\tau} \right),$$

with terminal velocity $v_t = mg/b$ and time constant $\tau = m/b$.

(b) $v = 0.95 v_t$ when $e^{-t/\tau} = 0.05$:

$$t = \tau \ln 20 \approx 3.0 \tau.$$

(c) **Red blood cell.** Its mass is

$$m = \rho V = 1100 \times \frac{4}{3} \pi (4.0 \times 10^{-6})^3 = 1100 \times 2.68 \times 10^{-16} = 2.95 \times 10^{-13} \text{ kg,}$$

and the drag constant is

$$b = 6\pi\eta r = 6\pi(1.2 \times 10^{-3})(4.0 \times 10^{-6}) = 9.05 \times 10^{-8} \text{ kg/s.}$$

$$\tau = \frac{m}{b} = \frac{2.95 \times 10^{-13}}{9.05 \times 10^{-8}} = 3.3 \times 10^{-6} \text{ s.}$$

The cell reaches terminal speed within about 10 μs . Compared with a test lasting an hour, it moves at terminal speed essentially all the time, so the assumption is excellent. For microscopic particles in viscous fluids, inertia is negligible. Biologists describe this as life at low Reynolds number.

P4.17

Rotating-frame analysis. In the hoop's frame, the bead at angle θ from the bottom is at distance $r = R \sin \theta$ from the axis. Three forces act on it:

- the weight mg , down;
- the normal force N from the hoop, along the radius of the hoop;
- the centrifugal force $m\omega^2 R \sin \theta$, horizontally outward from the axis.

The normal force has no component **along** the hoop (tangentially), so equilibrium requires the tangential components of the other two forces to balance. At angle θ , gravity's component along the hoop, pulling towards the bottom, is $mg \sin \theta$. The centrifugal force's component along the hoop, pushing away from the bottom, is $m\omega^2 R \sin \theta \cos \theta$. So

$$mg \sin \theta = m\omega^2 R \sin \theta \cos \theta.$$

Either $\sin \theta = 0$ (the bead sits at the bottom), or

$$\cos \theta = \frac{g}{\omega^2 R}.$$

Condition. A solution needs $\cos \theta < 1$, so $\omega^2 R > g$, that is, $\omega > \sqrt{g/R}$. Below this speed, the bottom is the only equilibrium position.

Numbers. $\omega = 120 \text{ rpm} = 120 \times 2\pi/60 = 12.6 \text{ rad/s}$, so $\omega^2 R = 157.9(0.20) = 31.6 \text{ m/s}^2$.

$$\cos \theta = \frac{9.8}{31.6} = 0.310 \Rightarrow \theta = 71.9^\circ.$$

Evaluate. This is a simple example of a *bifurcation*: as the spin rate passes $\sqrt{g/R}$, the bottom position becomes unstable and two new equilibria appear, one on each side. The same idea

appears in the centrifugal governors once used to regulate steam engines.

P4.18

(a) The coin speeds up along its circle, so it needs a **tangential** acceleration $\mathbf{a}_t = \alpha \mathbf{r}$ as well as the centripetal acceleration $\mathbf{a}_c = \omega^2 \mathbf{r}$. Static friction is the only horizontal force on the coin, so it must supply both. The friction vector therefore points partly forward (along the motion) and partly inward. It points straight at the centre only when $\alpha = 0$.

(b) **Slipping condition.** The coin slips when the total required friction reaches $\mu_s mg$:

$$m\sqrt{(\omega^2 r)^2 + (\alpha r)^2} = \mu_s mg.$$

With $\mu_s g = 3.92 \text{ m/s}^2$ and $\alpha r = 2.0(0.10) = 0.20 \text{ m/s}^2$:

$$(\omega^2 r)^2 = 3.92^2 - 0.20^2 = 15.37 - 0.04 = 15.33 \Rightarrow \omega^2 r = 3.915 \text{ m/s}^2.$$

$$\omega^2 = \frac{3.915}{0.10} = 39.15 \Rightarrow \omega = 6.26 \text{ rad/s}.$$

Time to reach this speed from rest:

$$t = \frac{\omega}{\alpha} = \frac{6.26}{2.0} = 3.13 \text{ s}.$$

Evaluate. The tangential part is small here (0.20 compared with 3.9 m/s²), so the coin slips at almost the same speed as it would on a steadily rotating turntable, $\omega = \sqrt{\mu_s g / r} = 6.26 \text{ rad/s}$. With a much larger α , the coin would slip much earlier. That is why a quickly started turntable flings coins off sooner.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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